

Development of a 3D HUD using a tunable bandpass filter for wavelength multiplexing

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Abstract — A 3D stereoscopic head-up display using a tunable bandpass filter to perform left and right image spectral separation is presented. Using a single filter reduces the size and the cost of the head-up display optical engine and enables each spectral band to be accurately tuned. Experiments performed on the first prototype demonstrate the ability to continuously tune the bandpass frequency on 30-nm range while keeping a 20-nm bandwidth. Such a system avoids the use of a bulky and costly rotating wheel and enables the use of holographic optical elements known to be wavelength selective.

Keywords — *head-up display, augmented reality, tunable filter, 3D display, spectral multiplexing, holographic optical elements, wavelength selectivity.*

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1 Objective

Head-up displays (HUD) have been used in aircraft for several decades. Initially designed for military applications as a means of targeting, they are now used to improve situational awareness, thanks to the ability to superimpose information in conformity with the external landscape. In a way, the HUD was the first application of what is now called augmented reality.

In order to reduce accommodation time, and to reduce parallax issues, images are collimated, so that symbols are perceived as being far away.

But the head projection unit able to generate a bright collimated image within a large user exit pupil and with the correct image quality is very bulky, and the associated complex optical engine is very expensive.

Direct projection on a transparent screen, associated with simulated collimation performed by a stereoscopic projector, is a promising solution.

In this paper, we present a novel compact optical engine for a stereoscopic HUD projector, using a tunable filter to perform the spectral multiplexing necessary for displaying left and right image.

2 Background

A HUD is a see-through system and requires an outstanding transmission rate.

Legacy 3D stereoscopic systems, using active shutter goggles or polarized goggles, cannot achieve the transmission rate of 70% required by the HUD standard.¹

An innovative solution, using spectral multiplexing on green-only image, was presented in document,² achieving a transmission rate of 75%.

This system requires two images to be displayed at two different wavelengths, by means of two filtered projectors and a pair of 3D glasses as shown on Fig. 1.

3 3D principle

A 3D system using polarized goggles cuts off at least 50% of the external luminance, due to light polarization.

An active shutter 3D system achieving more than 50% transmission of external luminance will require complex synchronization mechanism between projection source and temporal eye shuttering and will so become tricky and bulky on head.

The most suitable setup for a see-through system seems to be the wavelength multiplexing approach.³ By using a bandpass filter for each color and each eye, it is possible to achieve a transmission rate of 60% (and more if narrow band sources are used) while using only a light passive component on head (Fig. 2).

3.1 Spectral modulation for a head-up display

For an avionics HUD, displayed images are green, as requested by current Avionics standards.¹

So, spectral multiplexing only uses the color green for the goggles notches, with an outstanding transmission rate of 75%.

The principle presented removes only a small bandwidth of the visible spectrum, typically less than 30 nm of green at

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FIGURE 1 — Projection system with two projectors and 3D spectral glasses.²

520 nm and at 540 nm, resulting in a very high photopic transmission.

As these two notches are applied, respectively, to left and right eyes, the entire spectrum is visible by the two eyes, except for the notch wavelengths, only visible by one eye.

In order to achieve spectral multiplexing, it is possible to use two filtered projectors, but such a system is bulky and not compatible for short-range projection as required in aircraft or car cockpits.

The use of one short throw projector, used in temporal mode, is a preferred solution, but requires doubling the image rate.

3.2 Projector issue

Wavelength multiplexing in temporal mode requires the use of a segmented color wheel, with two bandpass areas.

Historically, this wheel was placed inside the projector, and the rotation was synchronized with the images. Such a system is known as Dolby 3D.⁴ But this system is bulky, and the size of the spinning wheel is not compatible with a vehicle projector.

An alternative solution involves modifying the internal optical path of the projector so as to pass alternately through a first colored filter and then a second colored filter.⁵ This type of technical solution has several disadvantages. It requires a complex optical system having two different light paths and two different colored filters.

3.3 Windshield issue

In this concept, the images for left and right eye are projected onto a screen located at a finite distance in front of the pilot or the driver. In order to avoid the installation within the cockpit of an additional optical component (with its mechanical interfaces, positioning and safety mechanisms), a preferred solution is to use the windshield as this projection screen.

The windshield screen must then present the following capabilities:

- It must allow see-through view of the external scene, with a transmission factor as high as possible, without affecting the resolution of the see-through image as seen by the pilot.
- When illuminated by a projector, each point of the screen has to reemit the energy received from the projector in a solid angle compatible with the required eye motion box dimensions, with a luminance compliant with the HUD contrast requirement. Practically, to ensure the legibility of the symbols presented by the HUD, the luminance level of the HUD image must reach at least 30% of the

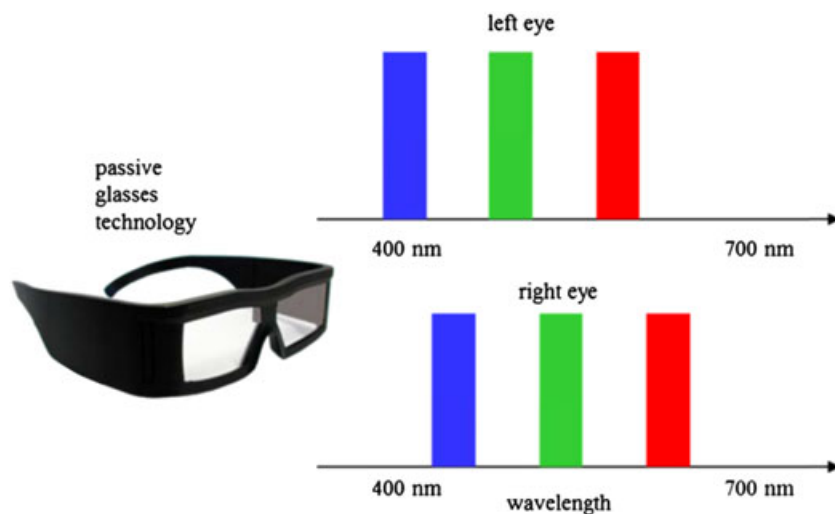


FIGURE 2 — Spectral separation principle: Left and right images are coded in spectrally independent wavelengths.³

luminance of the background as transmitted through the windshield.

One solution to improve the compromise between the required transmission and reflection properties of the windshield screen is obtained by working on the spectral dimension of light.

Typically, the external scene is illuminated by the sun on a wide spectral range, limited here to the visible domain by the pilot's physiological characteristics.

We can then take advantage of a narrow bandwidth illumination source for the HUD by adjusting the reflection efficiency of the screen to this specific narrow bandwidth.

As can be seen in Fig. 3, only a small portion of the spectrum is removed in transmission, providing a high see-through photopic transmission. On the other hand, when the HUD illumination spectrum is adapted with this reflection peak (both in position and width), then most of the energy coming from the projector will be reflected towards the pilot, with limited loss in transmission.

One remaining issue is to achieve the required “reflective–diffusive” function of the screen. This is done through the use of holographic optical elements (HOEs). As this is outside the subject of this paper, only a short description is provided hereafter. At each point on the screen, a specific optical function is created by way of reflection gratings, enabling the incoming light to be spread over the required solid angle at a given wavelength. This function is then recorded as a volume phase hologram on a photosensitive medium. At playback, the function will be activated for any light respecting the local Bragg condition between wavelength and angle of incidence (AOI). Away from the Bragg condition, the function is not active, and the screen is working as a passive transmissive component.

Typically, and for a high diffractive efficiency component, the spectral bandwidth of the HOE is proportional to the index modulation within the volume phase hologram. With the photopolymers available today, the spectral bandwidth varies between 8 and 10 nm.

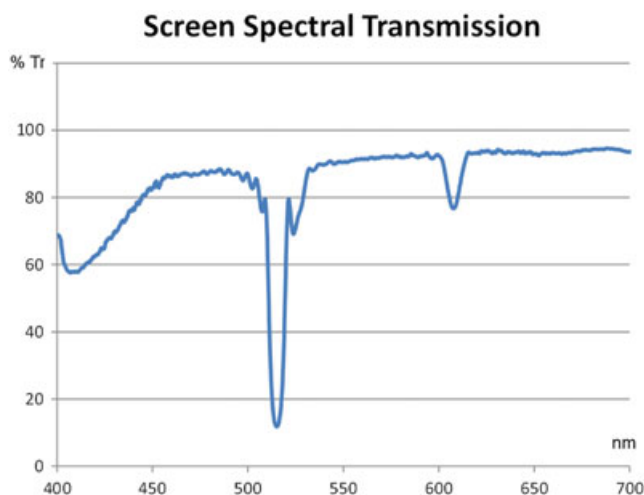


FIGURE 3 — Spectral transmission of the windshield screen using an HOE.

The peak wavelength position is determined by the holographic recording setup but can be marginally affected by the photopolymer shrinking factor (thickness variation between recording and playback). To maximize the final energy throughput, a capability to finely tune the incoming spectral bandwidth is highly desirable.

For this 3D HUD concept, the windshield screen is composed of two HOEs, one active for the right eye image, one for the left eye image.

4 Spectral modulator with tunable filter

Existing solutions use a segmented rotating wheel with two bandpass filters, or two separate bandpass filters fitted in front of two separate projectors.

These solutions require several hardware devices, increasing the system cost and complexity.

The proposed system⁵, shown on Fig. 4 uses only one bandpass filter (22) having the ability to finely tune the wavelength, from left eye set to right eye set. The imaging engine is composed with a reflective imager (20) and an illumination engine (21).

By using a rotating actuator, it is possible to select from the set of green images (I_G) by switching quickly and accurately between the two wavelengths, synchronously with the image rate.

4.1 Tunable filter

Bandpass filters are interferential filters: they transmit only a portion of the spectrum of the incoming light, as per user specification (central wavelength and spectral bandwidth at a defined AOI), and reflects the remainder of the incident light, resulting in spectral response as per Fig. 5.

Interference conditions form a relationship between the peak wavelength and the AOI on the filter. An increase in incidence induces a change in the filter spectral response, which can be approximated by a translation towards shorter wavelengths, i.e. “blue shift”, allowing the possibility to finely tune the frequency.⁶

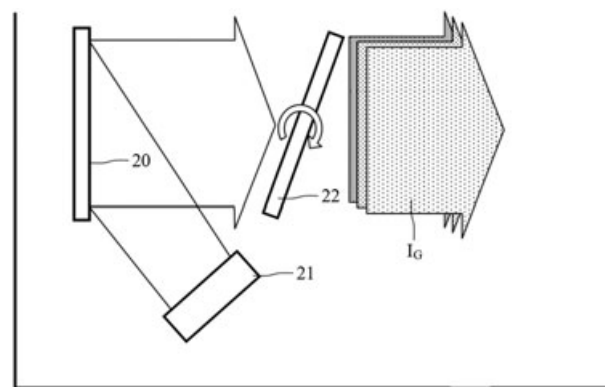


FIGURE 4 — System view: Tunable filter (22).⁵

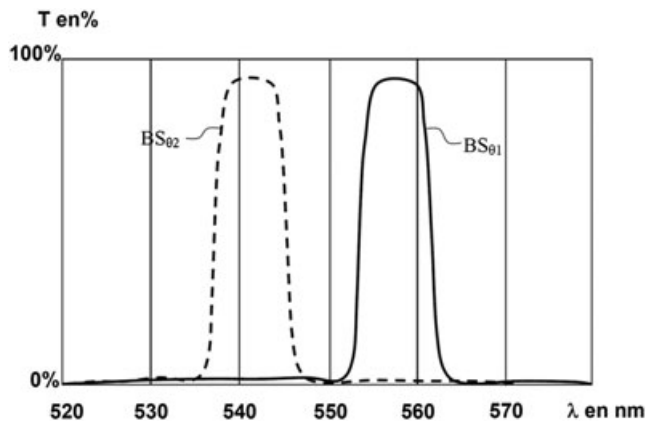


FIGURE 5 — Tunable filter bandpass per AOI (theoretical).⁵

This shift is not linear with incidence: small at low incidences, it becomes substantial at high incidences. Hence, a variation of incidence of 5° will have much stronger effects at 25° than at 0° .

4.2 Tilt angle actuator

Displaying stereoscopic images requires alternating the left and right images at a minimal frequency of 100 Hz. The tunable filter must then be rotated at this rate, with a cycle as near as possible to a square.

Electromagnetic rotating actuators (called galvo), and generally used on laser scanning system (Light show), could be a solution, but such devices are bulky and noisy, and difficult to integrate on an embedded system.

Paper⁷ presents such a system, using a piezoelectric actuator to tune the wavelength frequency. Whereas the dynamic characteristics of this system suit our application, the filter size needed for our HUD optical engine is too large for this kind of actuator.

Piezoelectric actuators were already successfully integrated into camera lens focusing system, with fast, accurate and silent operation. A dedicated piezoelectric rotating actuator, based on recent progress on mechatronics, is currently under development.

The main issue in stereoscopic projection using a temporal method is the crosstalk during the left and right transition. For example, with active LCD shutter glasses, images shall be blanked during LCD rise and fall time of each glasses, during 5 to 10 ms. This blanking time reduces the perceived luminance and the user experience.

In our system, the effective transition duration between the two states is limited at 2 ms, thanks to the high acceleration rate delivered by the piezoelectric actuator, and the low inertia of the filter. In addition, the two goggles notches present a wide spectral overlap (10 nm) around 532 nm (also providing a laser strike protection, as explained in document²), midway between the two operational positions. This overlap helps to reduce the need of image blanking and to remove transition crosstalk.

4.3 Light source

By principle, the tunable filter selects only one spectral part of the lighting source to display the wavelength selective image necessary for 3D display.

Our first application only needs green color, so a wide spectrum green source is required. As discharge lamps are less reliable, less efficient and bulky, LED is the most desirable light source, but classical green LEDs present (around 40-nm FWHM) a narrow spectrum, insufficient for our application.

The proposed solution is based on a blue LED illuminating a YAG phosphor, offering a wide bandwidth (larger than 80 nm) centered around 530 nm. In this case, the left part of the spectrum is used for the left eye image, and the right part for the right image.

4.4 First experiment and results

A tunable filter was developed with the following performance levels: bandpass width: 20 nm, centered wavelength for an AOI of 0° : 545 nm, centered wavelength for an AOI of 25° : 520 nm. The notch filters used for the goggles have a width of 30 nm, centered, respectively, at 520 and 540 nm.

The tunable filter was fitted on a goniometric table, in front of a DMD imager (See Fig. 6).⁸

The illumination source used was a high power projection LED, with a wide bandwidth of 100 nm centered at 530 nm. The output field of view of the optical engine was $18 \times 30^\circ$.

The goal of the test was to evaluate the 3D quality, particularly the efficiency of the couple formed by the notch filter (goggle) and the filtered image.

The expected result is an overlapping of the notch filter for the complete image and for the two wavelengths.

4.4.1 Tunable filter at 0°

The test performed was qualitative; the image was projected onto a white screen and seen through the two notch filters on the goggles.

In this configuration, the left notch filter (at 540 nm) totally blocks the image, and the right notch filter is totally transparent, as shown in Fig. 7.

4.4.2 Tunable filter at 25°

At 25° , the result was not perfect for the right eye, with an incomplete filtering of the image side, as shown in Fig. 8.

4.4.3 First experiment result analysis

This first result shows a partial filtering for the AOI of 25° .

In this configuration, the tunable filter is located in front of the projector lens and is affected by the projection angle, as shown in Fig. 9.

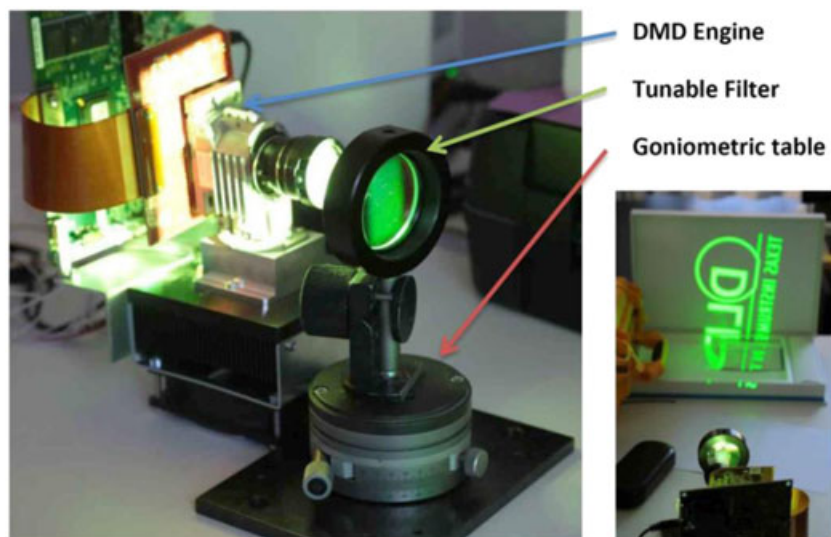


FIGURE 6 — Experimental setup.

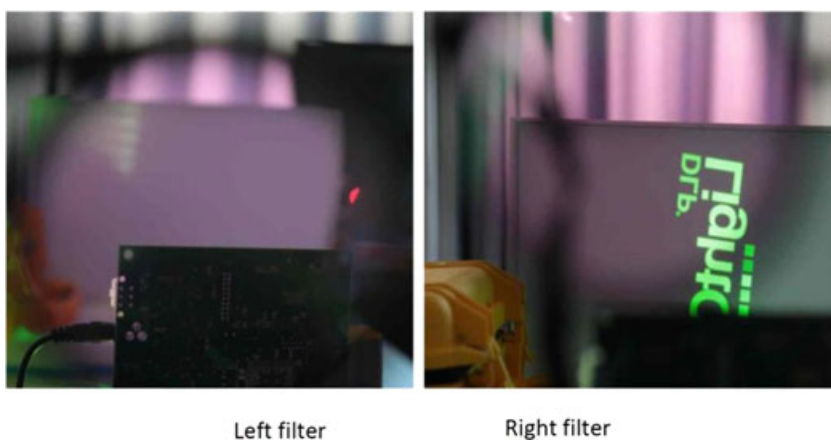


FIGURE 7 — Test result at 0°.

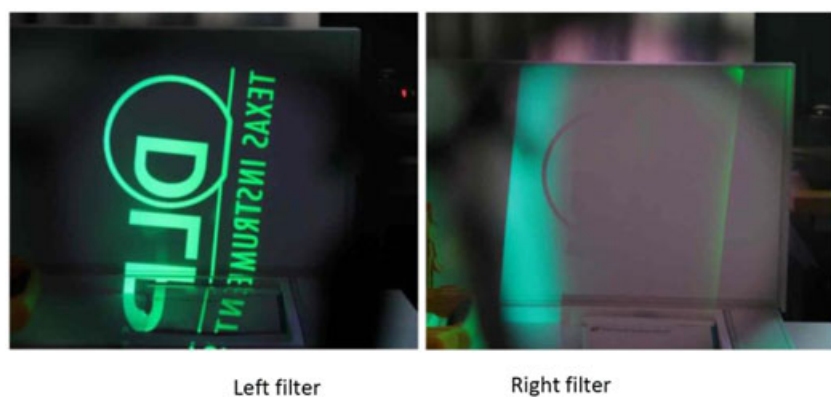


FIGURE 8 — Test result at 25°.

When the tunable filter is set at 0° for the center of the projected field of view, the effective incidence on the filter of the rays corresponding to the horizontal edges of the projected field of view is around 15°. Because of the blue shift (Fig. 8), the bandpass frequency is shifted from 545 to 539 nm. But with the goggle notch filter (540 nm $\pm$ 15),

the image is correctly overlapped by the goggle glass, as shown in Fig. 7—left filter.

When the tunable filter is set at 25° for the center of the projected field of view, the effective incidence on the filter of the rays corresponding to the horizontal edges of the projected field of view is 40° on the left side and 10° on the

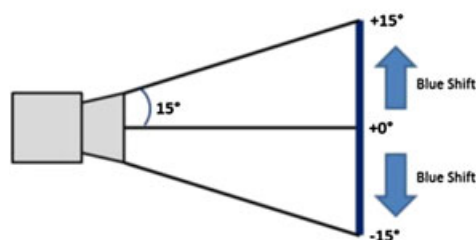


FIGURE 9 — Blue shift effect.

right side. The left side of the image is affected by a blue shift of 24 nm, while the right side is affected by a red shift of 15 nm. The total shift around 520 nm is now 39 nm, and the notch filter is unable to overlap the whole image, as shown in Fig. 8 for right image.

Using a front filter requires a narrow projection angle, with an impact on the throw ratio (ratio between the distance of the projector from the screen and the image width). For use on a vehicle, a short throw ratio is desirable. This is an issue for a wide-angle HUD.

4.5 Second experiment and results

For the first experiment, a tunable filter in front of the projector lens was used, with an issue on throw ratio.

To overcome this issue and provide good visual performance while keeping a low projection throw ratio, we designed a new lighting engine with the tunable filter integrated inside the optical path of an off-the-shelf projector.

Let us consider a typical white light projector, where white illumination is generated by mixing red, green and blue components. Such mixing is usually achieved with dielectric coatings, which will function with similar angular constraints (i.e. blue shift or red shift) as the tunable filter. To limit color variations in the projected image, the illumination system from LED sources to light integrator is designed so that the angular extension of the rays incident on the mixing components is minimal.

The modified projector was a more compact engine,⁹ based on a full HD DMD engine (0.47" Diag), and illuminated with a set of red, green and blue LED. The illumination engine was removed and modified to integrate the tunable filter and its actuator (0–30° angular course). Only the green LED was used, but its location was changed, to be in front of the tunable filter. In such projector, in order to increase its output efficiency, a collimating lens (one group in two elements) is placed between the LED and the tunable filter (Fig. 10), collecting high incidence rays emitted by the LED, and redirecting them towards the imager. When the LED is located at the object focal point of the lens, this also helps to reduce the angular spreading of the light incident on the tunable filter, strongly limiting blue shift effects through the tunable filter. The firmware

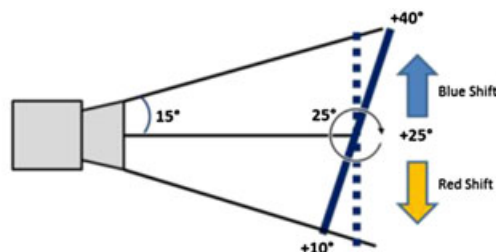


FIGURE 10 — Modified light engine.

in charge of color management was also modified to display a monochrome image.

As the piezoelectric motor is not yet available (under development), the wavelength tuning is performed by manually rotating a knob. The goal of this second experiment is to solve the previous issue and to measure the spectral performances for the two angular positions, corresponding to the two wavelengths of the HOE stack used as a transparent screen.

The illumination source used was a high power projection LED, with a wide bandwidth of 100 nm centered at 530 nm.

The expected result is an overlapping of the goggle notch filter for the complete image (without visible crosstalk), for the two wavelengths.

The reference image was the SID logo. The spectral luminance of three areas located on the center of each letter was measured, using a spectroradiometer.

4.5.1 Tunable filter at 0°

In this configuration, the left notch filter (at 540 nm) totally blocks the image, and the right notch filter is nearly totally transparent.

The peak wavelength was measured at 541 nm, with a FWHM of 18 nm. The three measured points give almost the same spectrum, with dispersion less than 1 nm.

On the blocking area (below 530 nm and above 555 nm), a better efficiency was noted on the center of the screen (letter I) (see Figs 11 and 12).

4.5.2 Tunable filter at 30°

In this configuration, the right notch filter (at 520 nm) nearly totally blocks the image, and the left notch filter is totally transparent.

The peak wavelength was measured at 518 nm, with a FWHM of 18 nm. The three measured points give the same spectrum.

On the blocking area (below 500 nm and above 535 nm), a better efficiency was noted on the left and on the center of the screen (letters S and I, see Fig. 11).

Regarding the cut-off efficiency, the slope at 30° was smoother than the slope at 0° (typically 18 nm to reach 100%) (see Fig. 12).

4.5.3 Spectral multiplexing efficiency/analysis of results

Figure 12 shows the result of these tests, at 0 and 30° of AOI, and for the three measured points. The goggle notch filters are also represented in dotted lines.

For each angular configuration (0° or 30°) of the bandpass filter (now located within the projector), the final illumination spectrum coming out of the projector is the integral sum of the spectral/angular response of the bandpass filter with the angular distribution of light at the filter location. Although this angular distribution is now much less than $\pm 15^\circ$ as when the filter was outside, the blue shift effect described in the Section on Tunable Filter still occurs, with stronger impact at 30° than at 0°. This translates in steeper profiles at 0° (slightly rounded top-hat shape) than at 30° (nearly triangular profile), with nearly similar FWHM, corresponding to the filter intrinsic spectral bandwidth.



FIGURE 11 — Measurement setup.

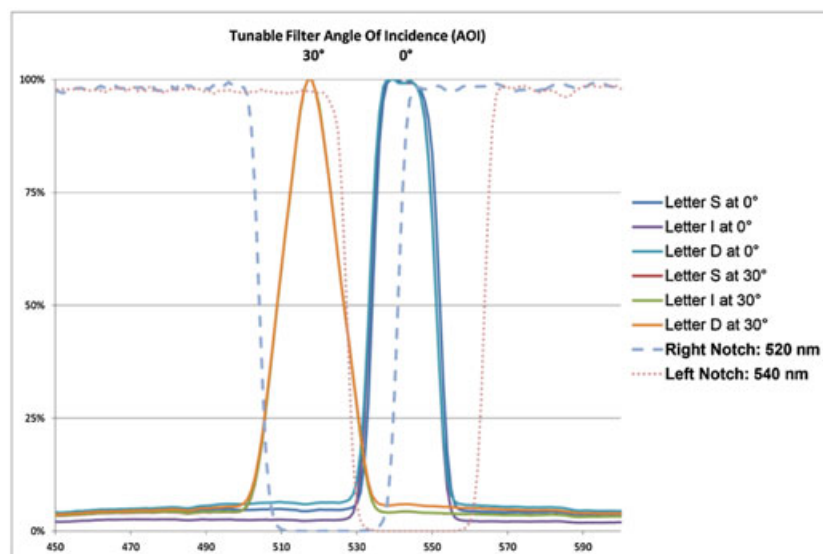


FIGURE 12 — Spectral measurements.

As could be expected, the variation of the illumination spectrum with the position in the image is now much reduced: the angular distribution of light on the DMD imager is by design nearly independent of the pixel position. This is validated by the very small spectral dispersion between the three sets of measures (letters S, I, D). In consequence, limitations of the notch blocking effect will mostly be felt over the whole field of view.

The right notch blocks most of the image at 30° of AOI, but a little part of the spectrum around 505 nm is incompletely attenuated as the bandpass slope is too low. However, rotating the filter by a few degrees (towards 25°) will be sufficient to shift the curve to the right and match with the notch filter. If necessary (this is not the case in our current system), it is possible to ensure complete blocking with the right notch by making it slightly wider to take into account the wider base of the illumination spectrum at 30° of AOI, at the expense of a slightly reduced outside world transmission for the right eye.

The left notch totally blocks the image at 0°, so no image crosstalk between left and right image is visible in stereoscopic vision.

Qualitative test, by wearing the spectral goggles shows a good separation between left and right images, without visible crosstalk.

5 Discussion

This setup was built for a proof of concept purpose. It shows the feasibility of using a tunable filter to perform a spectral multiplexing without using a bulky color wheel.

The front filter is the simplest solution, as it can be implemented without any modification to the projector. With a low throw ratio projector, this will induce an incomplete filtering on one eye, reducing the quality of the 3D vision. Alternatively, a higher throw ratio will provide better visual performance, but at the expense of a longer optical path from projector to screen, with potential space envelope issues when trying to install the complete system in the targeted vehicle.

To overcome this issue and provide good visual performance while keeping a low projection throw ratio, it is necessary to implement the tunable filter within the projection system, at a position where the angular extension of the rays will be minimal.

For a HUD system with only green imaging (Avionic application), it is then possible to remove the blue and red LED sources, and to replace the color mixing component with our tunable filter.

One of the installation constraints on a vehicle is the viewing direction, depending both on the pilot location and the windshield orientation. The use of interferential filter (3D goggle), or projection on HOE, needs to finely tune the image wavelength, to take into account the spectral shift of the rays versus the AOI. The tunable filter could optimize

the optical efficiency of the system, reduce crosstalk effect on stereoscopic images and improve the user experience.

For automotive applications, color is highly desirable, and today high-end HUDs are full color. On such device, color images are reflected by the windshield (used as a half-mirror). For a higher efficiency and transparency, we use two HOEs operating on two green wavelengths. Displaying color image is possible and will need two HOEs per color component (red, green, blue), one for each eye. The tunable bandpass filter will be more complex, with three bands (one for each primary color), as filters used for 3D projection,⁴ with higher constraints on the spectral position and bandwidth of each band, but still enabling the replacement of the bulky rotating wheel. Such solution will require the use of wide spectrum color sources. Today, compatible LED sources are only available in green for single chip components. For blue and red, the use of two bins per color is a possible solution.

6 Impact

A new HUD architecture was presented, using a tunable filter. The capability to accurately tune the wavelength of the two images for a stereoscopic projection is of great value, allowing the blue shift compensation of the goggle filter in case of wide viewing angle.

Additionally, if diffractive or HOEs need to be used, fine adjustment of the spectral source by means of the tunable filter is a great improvement, drastically reducing frequency crosstalk on 3D display. Even if this first study shows a monochrome application, a triple bandpass filter could be used to display a 3D image by multiplexing the two sets of red, green and blue colors.

Acknowledgments

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